

Developing a 3D interface for sound corpus manipulation in virtual environments

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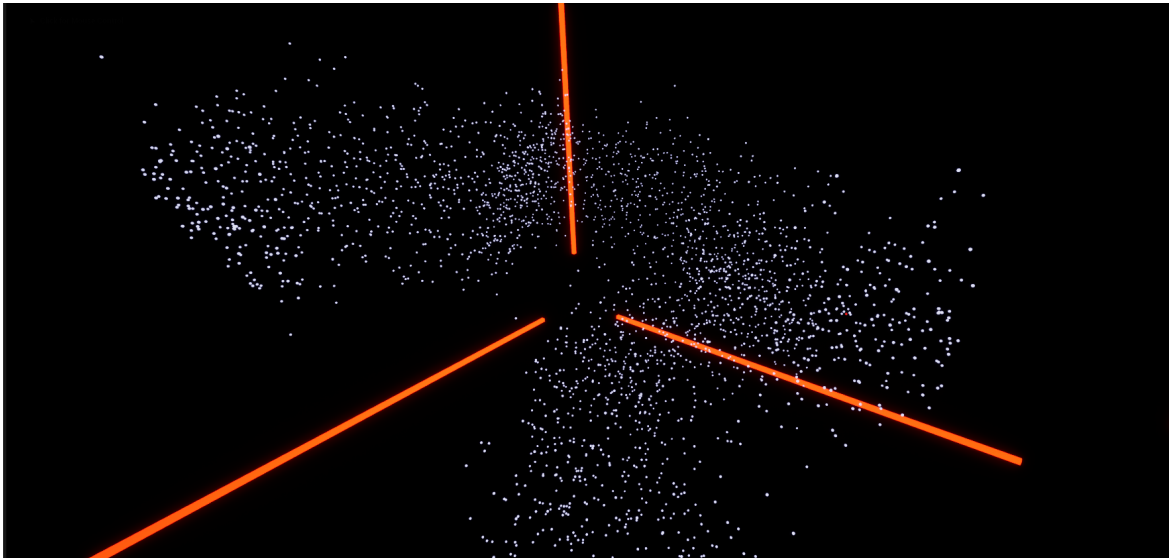


Figure 1: Samples from a sound corpus plotted in 3D space as points.

ABSTRACT

This paper describes the implementation of 3D Corpus Navigator, a virtual musical instrument that can be used to interact and reconfigure a collection of sounds in 3D space, with applications in sound design, music composition, and audiovisual performance. Using different sound descriptors to analyze a sound corpus, the results are plotted as a 3D point cloud where each point represents a sound sample. By turning the user into a virtual ‘playhead’, a single user can interact with the 3D point cloud by means of virtual navigation, playing back the sound samples as the user flies through the point cloud. The user can interact via a game-pad or mouse and keyboard. The virtual musical instrument was implemented using Unreal Engine and the FluCoMa library in Cycling 74’s Max/MSP with bi-directional Open Sound Control communication. This paper discusses how this implementation advances prior work of similar systems, its advantages, and limitations. Future work and improvements are outlined for new iterations of the system at the end of this paper.

Index Terms: Virtual environment, virtual navigation, concatenative synthesis, sound corpus.

1 INTRODUCTION

Multiple artists and scholars have developed digital musical instruments that map virtual navigation in virtual environments (VEs) to different musical parameters. Hamilton [8] modified the game

engine of a ‘First Person Shooter’ (FPS) to turn the game mechanics into sonic interactions for a multi-user electro-acoustic music performance. Berthaut et al. [4] created a collaborative multi-process musical instrument which relies on the interaction techniques used in FPS video games. Virtual navigation has been explored by Hamilton [9] as a procedural system to control timbre, amplitude and frequency of a sound synthesis algorithm based on the position of a flying virtual avatar relative to the landscape in the VE. Barri [1] thinks of 3D space as an audiovisual sequencer where the user can fly through rhythmic and drone-like audiovisual objects. The amplitude and spatialization of the sound objects is dependent on their euclidean distance to the user, simulating getting ‘close’ or ‘far’ from a sound source. Ciciliani [5] argues that VEs can be spaces for musical performance and also a medium to organize interactive sound sources and sound events contained in virtual topologies.

The 3D Corpus Navigator advances prior work by integrating Unreal Engine 5 and Max/MSP with bi-directional OSC [20] communication to create a FPS-style interface that relies on virtual navigation to control Corpus Based Concatenative Synthesis (CBCS). In the VE the user is turned into a virtual ‘playback head’ that can fly in 3D space. The user can interact with the 3D point cloud by flying through the different points, activating each sound sample upon contact. They can dynamically switch between different speed settings: slow, medium, and fast. The playhead’s position and velocity data is accessible to the user to customize interactive mappings that extend the sonic output of the sound corpus in Max/MSP. The goal of this project is to give users direct access to a virtual musical instrument with which they can perform CBCS using simple controllers to leverage interactions that are common to video games.

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2 RELATED WORK

CBCS is a technique that has been widely used in the domain of computer and electro-acoustic music [6, 15, 17, 18]. More recent frameworks propose adaptive mapping of sound collections for the development of data-driven musical interfaces [13]. Multiple interfaces and gestural controllers have been developed for real-time interactions with CBCS in musical performance [2, 14, 16, 22]. *Plumage* [10] is a 3D interactive system that uses virtual playheads to control CBCS. Sound grains are spread in space in the form of virtual feathers. It uses controllable playheads that can move in cyclical trajectories. The playheads have associated ‘triggers’ that orbit around them. The triggers activate the sound grains they come in contact with.

Immersive interactions with CBCS have also been explored using VR and AR systems. Zappi et al. [21] developed an immersive installation using inertial and marker-based tracking systems where users can interact with a sound corpus distributed in virtual space via hand gestures. Neupert [12] created a VR system using a low-cost phone based setup and a gesture tracker to immerse users in a virtual environment where they can perform concatenative synthesis on 3D point cloud through theremin inspired interactions. Halac and Addy [7] created an interactive VR experience that displays timbre descriptor data from a sound file as a 3D point cloud. Using handheld-controllers users can create different musical sequences and patterns from the point cloud. More recently, Berthaut [3] developed a mixed reality musical instrument using a Spatial Augmented-Reality display to project 3D textures onto physical space enabling users to control granular synthesis via hand gestures.

3 IMPLEMENTATION

The integration of Unreal Engine and Max/MSP supports a wide range of creative possibilities for interactive computer music performance and composition. Unreal Engine offers a robust interactive platform with enhanced graphics, while the FluCoMa library in Max/MSP offers a variety of tools for descriptor analysis, interactive sample playback, and live-processing given its configurability and scalability [19].

3.1 Max/MSP

The Max/MSP interface makes it simple to work with any sound collection owned by the user. Users can select different algorithms from the FluCoMa library to segment and analyze the sound corpus, and fine-tune the available parameters, which can be saved and recalled with the preset object. These parameters give flexibility to the user to determine how to segment the sound corpus, and the type of analysis to apply, determining the spatial arrangement of the 3D point cloud in the VE. The following paragraphs describe the different steps and sections of the Max/MSP interface that is shown in figure 2.

3.1.1 Segmentation, analysis and 2D plotting

The user can select between different onset detection algorithms to segment the corpus into samples. The user has to select two different analysis algorithms to apply to the sound samples, in order to plot the XY plane and Z axis respectively. For the XY plane users can choose to analyze the samples using MFCC, Spectral Shape or MelBands, while for the Z axis they can choose between Loudness, Pitch, or Chroma analysis.¹ All the datasets generated are normalized in this step. The XY plane is shown in a 2D representation for the user to have a previsualization before plotting the sound corpus in 3D space.

¹For more information on the different analysis algorithms available in FluCoMa refer to <https://learn.flucoma.org/reference/>

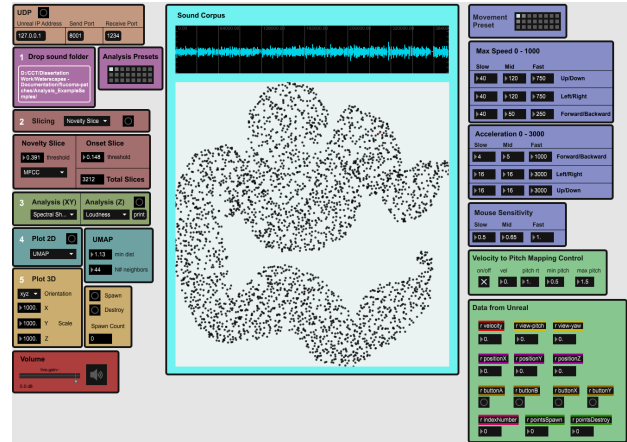


Figure 2: Max/MSP interface.

3.1.2 Plotting in 3D

Users can adjust the orientation and scaling of the XY plane + Z axis in Max/MSP before sending the position data for each sample to plot the 3D point cloud in the VE. The point cloud can be reoriented in all of its permutations, indicated in the form: XYZ, ZXY, YXZ, etc. The scale for each axis is determined separately.

3.1.3 Movement presets

In this section of the patch the user can adjust the maximum speed and acceleration parameters for the three different speed settings used for virtual navigation in 3D space. These settings can be saved and recalled in a separate preset object. For the mouse and keyboard configuration the user can adjust the mouse sensitivity associated with each speed setting.

3.1.4 Playback

The user can toggle between playing the samples at their original pitch, or mapping the playback speed to the real-time velocity of the virtual playhead in the VE. The mapping for the playback speed range can be adjusted between two octaves above and below the original pitch. The implementation is polyphonic allowing multiple samples to be triggered simultaneously and with overlap.

3.2 Virtual environment

There are multiple benefits to using a game development engine for the design of the VE. Unreal Engine 5 accelerates the process of assigning input mappings for control, and has an integrated OSC plug-in, allowing for communication with other music softwares like Max/MSP or Ableton Live.

In the VE, the user has a first-person view, with no virtual embodiment. They appear to be in a seemingly empty space, in front of an arrangement of edges that looks like a cube with disconnected vertices as seen in figure 1.

3.2.1 Movement controls

The ‘WASD’ keyboard keys are used to move forward/backwards and left/right, while the ‘Q’ and ‘E’ keys are used to move up and down respectively. The mouse is used to look around, which changes the direction of the forward vector. The user moves by default in speed setting 1, they can hold down the ‘shift’ key while moving to engage speed setting 2, and simultaneously hold down the ‘shift’ + ‘space’ key to engage speed setting 3.

Movement in the VE can also be controlled with a game-pad as shown in figure 3. This example is modeled after a Nintendo Switch

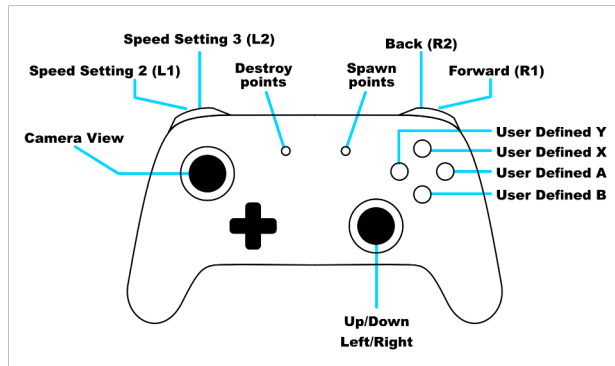


Figure 3: Game-pad controller mapping.

Pro Controller, but any generic game-pad with a similar layout can be used, including Xbox, or PlayStation controllers.

The left joystick is used to look around, while the right joystick is used for up/down and left/right movement. Forward and backwards movement is controlled with the right hand triggers ‘R1’ and ‘R2’ respectively. The left hand triggers are used to access the different speed settings: trigger ‘L1’ for setting 2, and trigger ‘L1’ + trigger ‘L2’ for setting 3.

The input data from the ‘user defined buttons’ are available to the user through the Max/MSP patch and can be used to select between different sound mappings to extend the sonic interactions as discussed in the latter section of this paper.

3.2.2 ‘Spawn’ and ‘Destroy’

The user can press the ‘enter’ key or the ‘start’ button on the game-pad to spawn the samples as virtual 3D points. The points are spawned at a rate of ten miliseconds, with the option to play their assigned sample in the process. This option can serve as a way to audibly survey the complete sound corpus while visualizing the process of populating the 3D space. Pressing the ‘backspace’ key or the ‘select’ button on a game-pad causes all points to be destroyed without any sample being triggered.

3.2.3 Sonic interaction

The user can interact with the 3D points via proximity, by passing through them as if the user were a flying virtual playhead. The collision box is set as a ‘trigger’, meaning that there is no physically simulated collisions with the points in 3D space. As the user clips through the points to trigger sound samples, the velocity registered at the moment of contact can be mapped to determine the playback speed of the sample. Users can create new sonic material through the real-time reconfiguration of a sound corpus based on their choices of trajectory and speed. The color of the 3D points change when triggered. This serves to give the user visual feedback of which points have been activated while leaving a visual trail of the movement enacted by the user.

4 DISCUSSION

4.1 Corpus Analysis and Scalability

Other CBCS systems based in FluCoMa such as *Mosaïque*² use descriptor analysis to look for self-similarity in the sound corpus and render a 3D visualization. *Plumage* uses fixed sound descriptors associated with each axis to plot grains of sound in 3D space. The 3D Corpus Navigator offers flexibility by giving users the option to select between the different analysis algorithms available in FluCoMa. The selection of algorithms together with the scaling and

orientation parameters for each of the three axis allow users to explore different spatial arrangements of one sound corpus.

4.2 Virtual Navigation

In *Plumage* the user can control the speed and trajectories of multiple virtual playheads as they fly through ‘grains’ of sound in a 3D environment. In this VE the user is looking at the playheads and grains from a third-person perspective. *Mosaïque* offers a 3D visualization of a sound corpus from a first person perspective, where users can change the camera distance and angle relative to the sound corpus rendered in 3D space. In *Mosaïque* users interact with the sound corpus by clicking and hovering the mouse cursor over the sound samples. The 3D Corpus Navigator sits between these two examples as it enables users to engage in creative exploration through the interaction with a sound corpus from a first person perspective in a VE. Although the system lacks the precision to select specific sound samples when compared to cursor based CBCS systems, the FPS-style virtual navigation leverages skills that are common to video games to offer a playful and learnable interaction paradigm [11].

4.3 Extended Sonic Interaction

Many CBCS systems apply different methods for the selection and rearrangement of samples from a sound corpus [6, 10, 14, 17, 22]. Some of the implementations in VR and AR use natural interactions to explore different methods to trigger and control points of a sound corpus[21], other systems extend the sonic capabilities by mapping gestural interactions to control the mix between granular synthesizers.[3]. The Max for Live version of *Mosaïque* offers similar possibilities where users can add audio effects at the end of the signal chain in Ableton Live. The 3D Corpus Navigator allows users to personalize and extend the sonic interactions beyond the real-time reconfiguration of sound samples via virtual navigation. The Max/MSP patch gives users access to incoming data from the VE via OSC, including the virtual playhead’s position, velocity, camera pitch and yaw, sample index, and controller input. This enables users to integrate their own audio processing effects in Max/MSP by mapping them directly to the interactions in the virtual environment. For example, the feedback of a simple delay effect can be mapped to the velocity of the virtual ‘playhead’, while the delay time can be mapped to the position in the Z axis, with different presets of this effect being selected from the ‘User Defined’ buttons on the controller or keyboard (3). An example of this can be seen in a screen recording of a live performance by the author where the live-processing of the audio using pitch and time based effects is mapped to the user’s velocity and position in the VE.³

4.4 Usability

Using Unreal Engine 5 and Max/MSP 8 facilitates the delivery and distribution of the system, making it accessible for artists and musicians to experiment with the system.⁴ The virtual environment can run on a Windows operating system. The Max/MSP collection file contains the analysis and playback patch and only requires users to have the runtime version of the software with the FluCoMa library installed to use at its most basic level.

4.5 Limitations

Compared to systems like *Plumage*, this first iteration of the 3D Corpus Navigator does not include sound spatialization, with the sound output being monophonic. The current version does not allow for dynamic reconfiguration of the points based on a new descriptor analysis. Users must destroy the points and re-start the

²<https://github.com/LFO-lab/Mosaïque>

³<https://matvilap.github.io/works/pointcloud.html>

⁴<https://github.com/matvilap/3D-corpus-navigator>

analysis process in the Max/MSP patch before re-spawning all the points with a new spatial arrangement.

5 FUTURE WORK

This first iteration serves as a proof of concept for the development of more versatile interactions and manipulations of a sound corpus in a 3D virtual environment. The next steps for the desktop version include developing networked multi-user capabilities, implementing spatialized sound, and the evaluation of the system through user studies to measure the instrument's intuitiveness, learnability, and efficiency [11] to develop new interaction modes for electronic music practitioners using this system.

The next iteration also contemplates a VR/AR implementation where users will be able to trigger the points using their hands through gesture tracking while exploring methods for distributing the 3D points in room-scale environments. More parameters can be used for representing the sound samples in virtual space, for example using the duration of the samples to determine the size of the 3D points, giving visual feedback for the user to differentiate the length between sound samples. The parameter settings could also be implemented as 3DUIs or pop up menus for the user to adjust analysis and playback settings without needing to take off the head-mounted display.

ACKNOWLEDGMENTS

Thanks to the developers of FluCoMa for making such a versatile tool, particularly Ted Moore for his help with learning how to use the library. And thanks to the Music Department at the University of Virginia together with the Jefferson Scholars Foundation for providing financial support for this research.

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